PHYSICAL SCIENCE - QUESTION BANK
SSC 100 DAYS ACTION PLAN

1. CHEMICAL REACTIONS AND EQUATIONS

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Explain the following with an example.
   i) Chemical combination ii) Chemical decomposition
   iii) Chemical displacement iv) Chemical double displacement

   (or)

   State and explain four types of chemical reactions with an example each.
2. Balance the following chemical equations
   i) $HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$
   ii) $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$
   iii) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
   iv) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$
3. Balance the following chemical equations
   i) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
   ii) $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$ (on heating)
   iii) $CH_4 + O_2 \rightarrow CO_2 + H_2O$
   iv) $Zn + HCl \rightarrow ZnCl_2 + H_2$
4. Write the balanced chemical equations for the following reactions.
   a) Calcium hydroxide + Carbon dioxide → Calcium carbonate + Water
   b) Zinc + Silver nitrate → Zinc nitrate + Silver
   c) Aluminium + Copper chloride → Aluminium chloride + Copper
   d) Barium chloride + Potassium sulphate → Barium sulphate + Potassium chloride
5. Write the balanced chemical equations for the following reactions.
   a) Potassium bromide(aq) + Barium iodide(aq) → Potassium iodide(aq) + Barium bromide(s)
   b) Zinc carbonate(s) → Zinc oxide(s) + Carbon dioxide(g)
   c) Hydrogen(g) + Chlorine(g) → Hydrogen chloride(g)
   d) Magnesium(s) + Hydrochloric acid(aq) → Magnesium chloride(aq) + Hydrogen(g)

1 MARK QUESTIONS

1. A magnesium ribbon is burnt in the presence of Oxygen to give Magnesium oxide. Rewrite the above reaction as Chemical equation.

2. Suggest one method to prevent corrosion.
3. Predict, Exhalation air is hotter than Inhalation air in respiration process.
4. Why do we apply paint on iron articles?
5. Why do we keep food in air tight containers?
6. Why does the colour of copper sulphate solution change when an iron nail is dipped on it?
7. Why is respiration considered an exothermic reaction?
8. What happens when iron nail is exposed in air or atmosphere? Predict
9. Oil and fat containing food items are flushed with nitrogen. Why?
10. Why do silver, gold and platinum not corrode in moist air?
11. Why is photosynthesis considered an endothermic reaction?
12. Why cannot we stir silver nitrate solution with copper spoon?
13. Why should a magnesium ribbon be cleaned before burning in air?
14. Name one observation that helps determine if a chemical reaction has occurred.
15. What is the new substance formed during a chemical reaction called?
16. Write two conditions of corrosion.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. What are the differences between displacement and double displacement reactions? Write equations for these reactions.
2. Explain the following terms with one example each.
   i) Corrosion ii) Rancidity
   iii) Oxidation iv) Reduction

1 MARK QUESTIONS

1. State the law of conservation of mass in a chemical reaction.
2. What is the common name for calcium oxide (CaO)?
3. What type of reaction occurs in the digestion of food in our body?
4. What type of reaction occurs when an iron nail is placed in copper sulphate solution?
5. What is the general term for the process where a metal is attacked by moisture or acids?
6. What is the main cause of damage to objects made of iron?
7. Name two effects of oxidation in daily life.
8. Name the gas that can be used for storage of fresh sample of chips for a long time.
9. What type of reaction takes place during respiration?

9. LIGHT–REFLECTION AND REFRACTION

LEVEL-1 : RISING STAR

4 MARKS QUESTIONS

1.

Material medium	Air	Ice	Rubby	Benzene
Refractive Index	1.0003	1.31	1.71	1.50

Observe the table and answer the following questions.

i) Which material medium light travels faster?

ii) In which material medium the speed of light is least?

iii) What is the speed of light in air?

iv) Calculate the speed of light in Benzene? (Speed of light in vacuum is 3×10^8 ms^{-1})

2.

Material medium	Air	Ice	Rubby	Benzene
Refractive Index	1.0003	1.31	1.71	1.50

Observe the table and answer the following questions.

i) Which material medium is optically rarer?

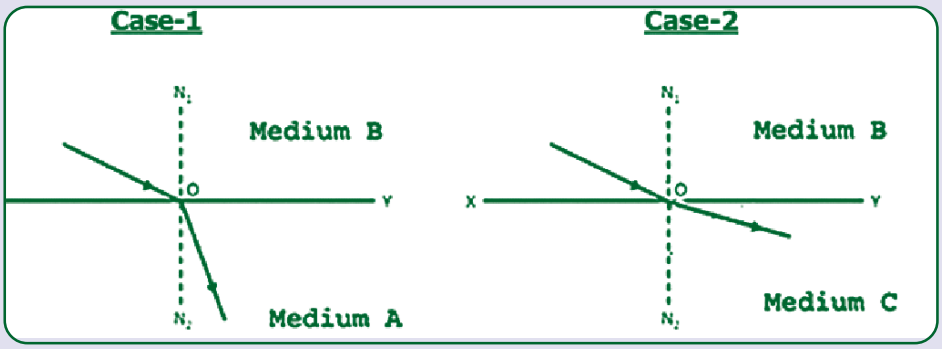
ii) Which material medium is optically denser?

iii) Write the relation between refractive index and speed of light in the medium?

iv) What is the SI unit of Refractive Index?

v) Arrange the above material media in the ascending order with respect to the speed of light.

3. Following diagrams show refraction of light in two cases. Answer the questions given below based on the diagrams given.



i) Which medium is optically rarer among A, B and C?

ii) Which medium is optically denser among A, B and C?

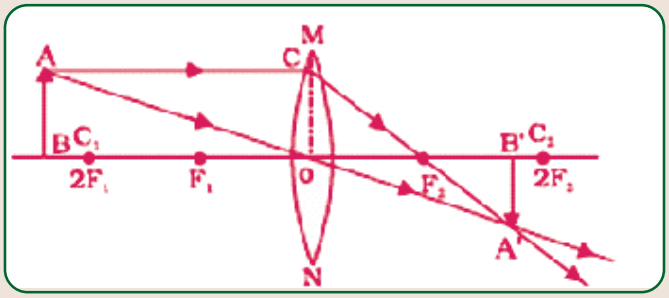
iii) Arrange A,B and C in ascending order with respect to speed of light.

iv) Arrange A,B and C in ascending order of their refractive indices.

4. Fill the table following, which is related to concave mirror.

Position of the Object	Position of the Image	Size of the Image	Nature of the Image
At infinity		Highly diminished, point-sized	
	At C		Real and inverted
Between C and F			Real and inverted
At F		Highly enlarged	

5.



Observe the ray diagram and answer the following questions.

i) Which lens is used in this ray diagram?

ii) Where is the position of the object?

iii) Where is the position of the image?

iv) What is the nature of the image?

v) If focal length of the lens is 10 cm, then what is the radius of curvature of this lens?

vi) Is magnification being less than 1 or greater than 1?

vii) If the height of the object is 10 cm at $2F_1$, then what is the height of the image?

6. Draw the ray diagrams of image formed when the object is placed infront of a bi-convex lens in the followingpositions.

a) Beyond $2F_1$ b) At $2F_1$

7. Draw the ray diagrams of image formed when the object is placed infront of a bi-convex lens in the followingpositions.

a) Between F_1 and $2F_1$ b) At F_1

8. Draw the ray diagrams of image formed when the object is placed infront of a concave mirror in the following positions.

a) Beyond C b) At C

9. Draw the ray diagrams of image formed when the object is placed infront of a concave mirror in the following positions.

   a) Between F and C b) At F

2 MARKS QUESTIONS

1. Predict and write about the world without lenses
2. Write any two daily life applications of lenses.
3. List two important uses of concave mirrors. (or)Write any two applications of concave mirrors.
4. List two important uses of convex mirrors.(or)Write any two applications of convex mirrors.
5. Write any two applications of mirrors.
6. Why do we prefer a convex mirror as a rear-view mirror in vehicles?
7. One-half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? Verify your answer (or) What happens to the image formed by a convex lens if its lower part is blackened ?

1 MARK QUESTIONS

1. The radius of curvature of a spherical mirror is given as 20 cm then determine it's focal length.
2. Find the Power of a Convex lens having a focal length of 50cm.
3. Find the focal length of a lens of power – 2.0 D?
4. A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens.

LEVEL-2 : SHINING STAR

4 MARKS QUESTIONS

1.

Material medium	Air	Water	Benzene	Diamond
Speed of light in the medium (m/s)	3×10^8	$9/4 \times 10^8$	2×10^8	$5/4 \times 10^8$

Observe the table and answer the following questions.

i) In which material medium light travels faster?

ii) In which material medium light travels lowest?

iii) Which material medium has the least refractive index ?

iv) Which material medium has the greatest refractive index ?

v) Calculate the refractive index of Benzene? (Speed of light in vacuum is 3×10^8 ms^{-1})

2.

Material medium	Air	Ice	Water	Kerosene	Benzene	Rock Salt	Diamond
Refractive Index	1.0003	1.31	1.31	1.44	1.50	1.54	2.42

Observe the table and answer the following questions.

i) What happens to the speed of light when light is passing from Water to Rock salt?

ii) Whether the refracted ray bends towards normal or away from the normal when light ray travelled from Benzene to Air?

iii) The refractive index of Diamond is 2.42. What is the meaning of this statement?

iv) What is the reason for refractive index of kerosene is more than the refractive index of water?

3. If radius of curvature of the mirror is double times of the focal length, then complete the following table

f (in cm)	R (in cm)
12	
24	
	15
	20

4. Fill the table following, that is relate to convex lens.

Position of the Object	**Position of the Image**	**Relative Size of the image**	**Nature of the image**
Beyond $2F_1$			Real and Inverted
	At $2F_2$	Same size	
Between F_1 and $2F_1$		Enlarged	
	Behind the lens		Virtual and Erected

5. Draw the ray diagrams of image formed when the object is placed infront of a bi-convex lens in the following positions.

a) At infinity b) Between F and Optical centre O

6. Draw the ray diagrams of image formed when the object is placed infront of a concave mirror in the following positions.

a) At infinity b) Between P and F

2 MARKS QUESTIONS

1. A ray of light travelling in air enters obliquely into water. Predict and write weather that light ray bends towards the normal or away from the normal? Why?
2. What happens to the image formed by a convex lens if its lower part is blackened?
3. The magnification produced by plane mirror is +1. What does this mean?
4. A ray passing through the centre of curvature of a concave mirror, after reflection, is reflected back along the same path.Why?
5. If A, B are optical medium of their refractive indices are nearly same, then light travel from A to B, What happens? (or) What happens to a ray of light when it travels from one medium to another medium having equal refractive indices?
6. If you want to see an enlarged image of your face, which type of mirror will you see? Where will you place your face?
7. What happens when a ray of light strikes the surface of separation between the two media at right angle?
8. What happens to a ray of light that passes through the principal focus of a concave mirror after reflection?
9. What happens to a light ray travels from denser medium to rarer medium?

1 MARK QUESTIONS

1. Find the focal length of a convex mirror whose radius of curvature is 32 cm.
2. Find the power of a concave lens of focal length 2 m.
3. The refractive index of diamond is 2.42. What is the meaning of this statement?
4. A lens has a power of +2.0 D. What type of lens is it?
5. The power of a lens is - 4.0 D. What is the nature of the lens?
6. If the magnification of a lens is -1, what is the nature of the image?

2. ACIDS, BASES AND SALTS

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Write an activity to show that the reaction of acids with metals. (or)
   Prove with activity when acids react with metals gives hydrogen gas.
2. Write an activity to show that metal carbonates and metal hydrogen carbonates react with acids. (or) Show that the reaction of Carbonates and Hydrogen carbonates with acids produces carbon dioxide gas
3. Compounds such as alcohols and glucose also contain hydrogen but are not categorised as acids. Describe an Activity to prove it. (or) Write an activity to show that acid solution in water conducts electricity.

4 MARKS QUESTIONS

1. Draw the diagram which shows that acid solution in water conducts electricity.
2. Draw a diagram of arrangement of apparatus for the reaction of acids with metals (or) Draw the diagram that showing the reaction of zinc granules with dil.HC*l*.
3. Draw a diagram of passing carbon dioxide gas through calcium hydroxide solution when metal carbonates or metal hydrogen carbonates react with acids.
4. Give important uses of washing soda (or) Write the uses of washing soda.
5. Give important uses of baking soda (or) Write the uses of baking soda.
6. Give important uses of Bleaching powder(or) Write the uses of Bleaching powder.
7. What are the applications of pH in daily life.

2 MARKS QUESTIONS

1. What happens if curd or sour substances kept in brass or copper vessels?
2. A milkman adds a very amount of baking soda to fresh milk.Why does shift the pH of the fresh milk 6 to slightly alkaline?
3. A milkman adds a very small amount of baking soda to fresh milk.Why does this milk take a long time to set as curd?
4. Fresh milk has a pH of 6. How do you think the pH will change as it turns into curd? Explain your answer.

5. Plaster of Paris should be stored in a moisture-proof container. Explain why?
6. What happens if a solution of sodium hydrogen carbonate is heated?
7. Why does distilled water not conduct electricity, whereas rain water does?

1 MARK QUESTIONS

1. Give example for an acid.
2. Give example for a base.
3. Give example for salt.
4. Give any one example for neutralisation reaction ?
5. What is the common name of the compound $CaOCl_2$?
6. The pH value of a solution is 10. What is its colour in the presence of methyl orange indicator?
7. What is the colour of phenolphthalein in a basic solution?
8. What happens when carbon dioxide is passed through lime water?

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Describe an activity to show the water of crystallisation.

4 MARKS QUESTIONS

1. Draw a diagram showing the variation of pH with the change in concentration of H^+ and OH^-

4 MARKS QUESTIONS

1. Give important uses of Plaster of Paris (or)Write the uses of Plaster of Paris.

2 MARKS QUESTIONS

1. Why is it recommended to add acid to water and not water to acid while diluting a concentrated acid?
2. How does using baking soda in a cake better make it soft and spongy?

3. What happens when water is added to quick lime?
4. How does the flow of acid rain into a river affect the aquatic life? (or) How does the flow of acid rain into a river make the survival of aquatic life in a river difficult ?
5. Why do acids not show acidic behaviour in the absence of water?

1 MARK QUESTIONS

1. Write the bleaching powder formula.
2. Write the Plaster of Paris formula.
3. Write the gypsum formula.
4. Name the substance which on treatment with chlorine yields bleaching powder.
5. Which gas is released on the reaction of zinc granules with dilute sulphuric acid. (or) Which gas is produced when acids react with metals?
6. Which gas evolves, when metal carbonate or metal hydrogen carbonate react with acids
7. What is the range of pH scale?
8. What is pH value of a neutral solution?
9. What types of medicines is used for treating indigestion?
10. Write an equation for the reaction between acid and base.
11. What happens when acids react with metals?
12. Name the sodium compound which is used for removal of permanent hardness in hard water.
13. What does a pH value less than 7 indicate?
14. Why are toothpastes generally basic?
15. What happens when Plaster of Paris is mixed with water?
16. How many molecules of water of crystallisation are there in Washing soda crystals.

10. HUMAN EYE AND THE COLOURFUL WORLD

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Explain, how do you correct the eye defect Myopia with a suitable diagram. (or)

   What is Myopia? How do you correct the eye defect Myopia? (or)

   Manoj is not able to see the letters clearly far from him. Identify the eye defect he has been suffering from and how can you rectify it? Explain.

2. Explain the correction of the eye defect Hypermetropia with a suitable diagram. (or)

   What is Hypermetropia? How do you correct the eye defect Hypermetropia? (or)

   Manoj is not able to see the letters clearly near from him. Identify the eye defect he has been suffering from and how can you rectify it? Explain.

1 MARK QUESTIONS

1. The change in focal length of eye lens is caused by the action of the _____
2. The least distance of distinct vision for a young adult with normal vision is about _____
3. The change in the curvature of the eye lens can thus change its _____
4. The minimum distance, at which objects can be seen most distinctly without strain is called _____
5. The human eye forms the image of an object at its _____
6. The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to _____
7. The ability of eye lens to adjust its focal length is called _____
8. The splitting of white light into its component colours is called _____
9. Vasavi cannot read the newspaper clearly. What type of eye defect she?

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Explain the following

   a) Scattering o light b) Dispersion of light

   c) Presbyopia d) Power of accommodation of eye lens

2. Explain the following.

   a) Ciliary muscles b) Atmospheric refraction

1 MARK QUESTIONS

1. The condition of the crystalline lens of old age people becomes milky and cloudy is called _____
2. Myopia is also known as _____
3. What is the other name of near sightedness?
4. How is Myopia or near sightedness corrected?
5. Hypermetropia is also known as_____
6. What is the other name of long sightedness?
7. How is Hypermetropia or long sightedness corrected?
8. What type of image formed by the eye lens?
9. A rainbow is formed due to _____
10. Twinkling of stars due to _____
11. The blue colour of sky due to _____
12. The phenomenon of scattering of light by the colloidal particles gives rise to _____
13. What is diameter of human eye?
14. Danger signals lights are red in clour. Why?

3. METALS AND NON–METALS

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Explain the experimental procedure to investigate the conditions under which iron rusts. (or) Suggest an activity to prove that the presence of air and water are essential for corrosion. Explain the procedure
2. How do metals react with solutions of other metal salts? Describe an activity. (or) Write an activity to show that high reactive metals displaces low reactive metals from their compounds.
3. Explain an activity to show the metals are good conductors of heat.

4 MARKS QUESTIONS

1. Draw a neat diagram to show the metals are good conductors of electricity.
2. Draw a neat diagram to show the action of steam on a metal (or) draw a neat diagram to show the metal reacts with water.
3. Write any four uses of metals.
4. Write any four uses of Non-metals.
5. Write any two advantages of avoiding corrosion and thermite process.

1 MARK QUESTIONS

1. Propose a method to extract a highly reactive metal from its ore?
2. Why oxides of high reactive metals cannot be reduced by carbon.
3. What happens to silver articles when exposed to moist air?
4. What happens to the insoluble impurities during the electrolytic refining of any metal?
5. Why pure gold is not used for making jewellery?
6. What happens to potassium and sodium if they are kept in open (or)
   Why potassium and sodium are immersed in kerosene?
7. Why do ionic compounds have high melting points?
8. What happens when magnesium is heated to its ignition temperature?
9. Name any one metal which displaces hydrogen from HNO_3.
10. What happens to the blue colour of copper sulphate solution when an iron nail is dipped in it?
11. Which one of the following is a non-metal?

    A) Magnesium B) Gold C) Sulphur D) Silver

12. Which of the following pairs will give displacement reactions?
    A) NaC*l* solution and copper metal B) $MgCl_2$ solution and aluminium metal
    C) $FeSO_4$ solution and silver metal D) $AgNO_3$ solution and copper metal.
13. Which of the following methods is suitable for preventing an iron frying pan from rusting?
    A) Applying grease B) Applying paint
    C) Applying a coating of zinc D) All of the above.
14. An element reacts with oxygen to give a compound with a high melting point. This compound is also soluble in water. The element is likely to be
    A) calcium B) carbon C) silicon D) iron.
15. Food cans are coated with tin and not with zinc because
    A) zinc is costlier than tin. B) zinc has a higher melting point than tin.
    C) zinc is more reactive than tin. D) zinc is less reactive than tin.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Describe an activity on metals react with water (Action of steam on a metal)

4 MARKS QUESTIONS

1. Draw a neat diagram to show that high reactive metals displace low reactive metals from their compounds. (or) Draw a neat diagram to show that metals react with solutions of other metal salts.
2. Draw a neat diagram to show the electric conductivity of a salt solution.
3. Draw a neat diagram to show the electronic refining of copper.
4. Write any four uses of refining of metals.
5. Explain, how each of these observations helps the student conclude that the substance is an ionic compound (Properties of Ionic Compounds).

1 MARK QUESTIONS

1. Give an example of a metal which is a liquid at room temperature.
2. Give an example of a metal which can be easily cut with a knife.
3. Give an example of a metal which is the best conductor of heat.
4. Give an example of a metal which is a poor conductor of heat.
5. Write any one physical property of metals.
6. What are amphoteric oxides?

7. Write any one of ways or method to prevent the rusting of iron or corrosion.
8. What type of oxides are formed when non-metals combine with oxygen?
9. Which metals do not corrode easily?
10. Write any one general properties of ionic compounds?
11. Name two metals which are found in nature in the free state.
12. What chemical process is used for obtaining a metal from its oxide?
13. Give one example of amphoteric oxides?
14. Name a non-metal which is lustrous.
15. Name two metals both of which are very ductile as well as very malleable.
16. Which metal is the most reactive according to the activity series table?
17. Name the least reactive metal shown in the activity series table.
18. Which of the following are not ionic compounds

    i) $NaCl$ ii) CCl_4 iii) KCl iv) HCl

    a) i) and ii) b) ii) and iii) c) ii) and iv) d) iii) and iv)
19. Aqua regia means

    a) 3:1 of Conc. HCl, Conc. HNO_3 b) 1:3 of Conc. HCl, Conc. HNO_3

    c) 3:1 of Dil. HCl, Dil. HNO_3 d) 1:3 of Dil. HCl, Dil. HNO_3
20. Branze is an alloy of __

    a) Cu and Zn b) Al and Sn c) Cu, Sn and Zn d) Cu and Sn
21. The reducing agent in thermite process is ____

    a) Al b) Mg c) Fe d) Si
22. Arranged the metals Fe, Na, Ag and Zn in increasing order of Reactive series.
23. Give any two examples of non-metals.

11. ELECTRICITY

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Deduce the expression for the equivalent resistance of three resistors connected in parallel in an electric circuit. (or) Explain the derivation $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$.
2. Deduce the expression for the equivalent resistance of three resistors connected in series in an electric circuit. (or) Explain derivation of $R = R_1 + R_2 + R_3$
3. Explain the following
   i) Electric current ii) Potential difference
   iii) Ohm's law iv) Electric power
4. Explain the following
   i) Fuse wire ii) Joule law of heat
   iii) Resistance iv) Resistivity
5. Write the S.I Unit of the following
   a) Electric Charge b) Electric Current
   c) Potential difference d) Resistance
   e) Resistivity f) Electric power
   g) Commercial unit of Electric Energy h) Heat

2 MARKS QUESTIONS

1. Predict and write why the series arrangement is not used for domestic circuits?
2. Why is the tungsten used almost exclusively for filament of electric lamps?
3. How does the resistance of a wire vary with its area of cross-section?
4. Why are copper and aluminium wires usually employed for electricity transmission?
5. Why should we connect electric appliances in parallel in a household circuit? What happens if they are connected in series?
6. Why are coils of electric toasters and electric irons made of an alloy rather than a pure metal?
7. Why does the cord of an electric heater not glow while the heating element does?

1 MARK QUESTIONS

1. Draw the symbol of an electric cell.
2. Draw the symbol of a battery or combination of cells
3. Draw the symbol of Plug key or switch
4. Draw the symbol of an ammeter.
5. Draw the symbol of a voltmeter.
6. Draw the symbol of the electric bulb.
7. Draw the symbol of a resistor of resistance.

8. Draw the symbol of rheostat or variable resistance.
9. Draw the V (Potential difference) –I (Current) graph for Ohm's law.
10. 2Ω, 4Ω resistors are connected in series. What will be the resultant resistance?
11. Two resistors 3Ω, 6Ω are connected in parallel. What will be the resultant resistance?
12. 1 KWH how many joules? (or) How many joules are there in one kilowatt hour?
13. How is an ammeter connected in a circuit? (or) How is an ammeter connected in a circuit to measure current flowing through it?
14. How is voltmeter connected in a circuit? (or) How is voltmeter connect in a circuit to measure potential difference across it ?

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Write the differences between resistance and resistivity.
2. Derive $H = I^2Rt$

2 MARKS QUESTIONS

1. What happens to the resistivity of a conductor if its length is doubled?
2. What happens to the resistance of conductor if its length is double and area of cross-section is halved?
3. What happens to the potential difference across a combination of resistors in series?
4. Why are alloys commonly used in electrical heating devices?
5. What would happen to the current if the potential difference across a component is halved while its resistance remains constant?
6. What happens to the current in a series circuit?
7. What is effective resistance of parallel combination of two 10Ω resistors?
8. What is the working principles of electric fuse?

1 MARK QUESTIONS

1. On what factors does the resistance of a conductor depend?
2. How can we measure the potential difference across the ends of a conductor?
3. What determines the rate at which energy is delivered by a current?
4. Calculate the number of electrons constituting one coulomb of charge. (or) How many electrons are contained in one coulomb of charge?
5. Which device can protect the home appliances from the high current ?
6. Which metal is used almost exclusively for filaments of electric bulbs?

7. What is the function of a fuse in an electric circuit?
8. What is the lowest total resistance possible with four coils of 4Ω, 8Ω, 12Ω, 24Ω
9. What is the highest total resistance possible with four coils of 4Ω, 8Ω, 12Ω, 24Ω?
10. What will be the resistance of a new wire if the original wire of 4Ω is made twice as thick?
11. How much energy is given to each coulomb of charge passing through a 6V battery?
12. Write the formula for electric current in terms of charge and time.
13. Write formula for potential difference.
14. Write the formula for Ohm's law?
15. State the expression for the equivalent resistance of resistors in series.
16. State the expression for the equivalent resistance of resistors in parallel.
17. Write a formula for electric power in terms of voltage and current.
18. Which material is the best conductor of electricity?
19. What are the factors on which the heat produced in a wire depends?
20. When do we say that the resistors are connected in series?
21. When do we say that the resistors are connected in parallel?
22. If two resistors of 6Ω and 12Ω were given to you, then how do you connect them to get 4Ω as resultant resistance.
23. How can three resistors of resistances 2Ω, 3Ω and 6Ω be connected to give a total resistance of 11Ω ?
24. How can three resistors of resistances 2Ω, 3Ω and 6Ω be connected to give a total resistance of 1Ω ?
25. Which is having more resistance a 220 V, 100 W bulb or a 220 V, 60 W bulb?
26. How many electrons continent one coulomb of charge?
27. Which of the following terms does not represent electric power in the circuit
    a) I^2R b) IR^2 c) VI d) V^2/R
28. Fuse is always connected in _____ the household appliances.
29. Which property of the conductor that resistance the flow of electric charge?
30. What is the function of rheostat in the electric circuit?

4. CARBON AND ITS COMPOUNDS

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Complete the following table.

No. of carbon in Hydro carbon	Alkane	Alkene	Alkyne
3	C_3H_8		
4	C_4H_8		
5	C_5H_8		
6	C_6H_{12}		

2. Explain the cleaning action of soap. (or) Explain the mechanism of the cleaning action of soaps.
3. Complete the following table.

Functional group	Suffix	Formula of functional group	Example
Alcohol			CH_3OH
Aldehyde	- al		
		- CO-	CH_3COCH_3
	- oic acid		CH_3COOH

2 MARKS QUESTIONS

1. Explain combustion with an example.
2. What are the two properties of carbon which lead to the huge number of carbon compounds we see around us?
3. What are hydrocarbons? Classify them into three main types based on bonding.
4. Give one example of saturated and unsaturated compounds
5. Write any two properties of Ethanol (Alcohol).
6. Write any two properties of Ethanoic Acid.
7. Explain esterification reaction with an example
8. What is a homologous series? Explain with an example.

1 MARK QUESTIONS

1. Which of the flowing hydrocarbon undergoes addition reaction?

   A) C_2H_6 B) C_3H_8 C) CH_4 D) C_3H_6
2. Which of the flowing hydrocarbon undergoes substitution reaction?

   A) C_2H_4 B) C_5H_{10} C) C_4H_{10} D) C_3H_6

3. Identify the alkane

   A) C_2H_4 B) C_5H_{10} C) CH_4 D) C_2H_2

4. Formula of Alcohol functional group is

   A) – OH B) – CHO C) – CO – D) – COOH

5. Formula of Aldehyde functional group is

   A) – OH B) – CHO C) – CO – D) – COOH

6. Formula of Ketone functional group is

   A) – OH B) – CHO C) – CO – D) – COOH

7. Formula of Carboxylic acid functional group is

   A) – OH B) – CHO C) – CO – D) – COOH

8. What are compounds containing only carbon and hydrogen called?
9. What are groups that confer specific properties to a compound called?
10. What is a series of compounds with the same functional group called?
11. Write the formula of first member of the homologous series to which C_5H_{10} belongs.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Explain the following

   i) Homologous series ii) Substitution reaction

2. How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties? (or)

   How would you distinguish experimentally between a alcohol and a carboxylic acid?

3. Explain cleansing action of soap.
4. Write the differences between soaps and detergents.

2 MARKS QUESTIONS

1. Differentiate between an alcohol and carboxylic acid chemically.
2. Explain how carbon's tetravalency contributes to its versatile nature.
3. How is ethanoic acid prepared commercially?
4. Explain the nature of the covalent bond using the bond formation in CH_3Cl.
5. Explain the formation of scum when hard water is treated with soap.
6. Why does micelle formation take place when soap is added to water?
7. Why are carbon and its compounds used as fuels for most applications?

1 MARK QUESTIONS

1. Ethane, with the molecular formula C_2H_6 has
   A) 6 covalent bonds B) 7 covalent bonds
   C) 8 covalent bonds D) 9 covalent bonds
2. Butanone is a four-carbon compound with the functional group
   A) carboxylic acid B) aldehyde
   C) ketone D) alcohol
3. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that
   A) the food is not cooked completely
   B) the fuel is not burning completely
   C) the fuel is wet
   D) the fuel is burning completely
4. The by-product of soap is
   A) Isoprene B) Glycerol C) Butene D) Ethylene glycol
5. Vinegar is a solution of
   A) 30% – 40% acetic acid in alcohol
   B) 5% – 8% acetic acid in alcohol
   C) 5% – 8% acetic acid in water
   D) 15% -20% acetic acid in water
6. How many carbon atoms are present in one molecule of ethanol?
   A) One B) Two C) Three D) Four
7. 100% pure ethanol is called
   A) Rectified spirit B) Absolute alcohol
   C) Denatured alcohol D) Power alcohol
8. What does carbon burn in oxygen to produce?
9. What reaction is used in the hydrogenation of vegetable oils?
10. What is the general formula for alkenes?
11. What type of reaction occurs when chlorine is added to hydrocarbons in sunlight?
12. What is the common name for ethanol?
13. What gas evolves when alcohols react with sodium?
14. What is the common name for ethanoic acid?
15. What reaction forms esters?
16. What is the valency of carbon?
17. Would you be able to check if water is hard by using a detergent?
18. How many structural isomers can you draw for pentane?
19. Why is the conversion of ethanol to ethanoic acid an oxidation reaction?

12. MAGNETIC EFFECTS OF ELECTRIC CURRENT

LEVEL-1 : RISING STAR

8 MARKS QUESTIONS

1. Explain the procedure to show that compass needle is deflected on passing an electric current through a metallic conductor (Oersted's experiment) (or) How can it be shown with the help of an activity that a magnetic field is produced around a current carrying wire? (or) Describe with the help of a diagram an activity to show that a current carrying wire behaves like a magnet.
2. Describe an activity on force experienced by a current-carrying conductor placed in a magnetic field. (or) Describe with the help of a diagram an activity to show that a current-carrying conductor experiences aforce when placed in a magnetic field.
3. Describe with the help of an activity to draw the magnetic field line around the bar magnet.

1 MARK QUESTIONS

1. State Right- hand thumb rule.
2. State Maxwell's Cork-Screw rule.
3. State Fleming's left- hand rule.
4. Write Oersted's law.

LEVEL-2 : SHINING STAR

8 MARKS QUESTIONS

1. Describe an activity to show the direction of magnetic field lines, produced by a current-carrying conductor.
2. Write an activity to show the shape and direction of the magnetic field line around a current carrying straight conduct.

1 MARK QUESTIONS

1. What is meant by magnetic field?
2. What is uniform magnetic field?
3. What is Solenoid?
4. What are the uses of fuse?
5. What is the direction of the magnetic field inside the bar magnet?
6. What is overload?
7. Choose the correct option. The magnetic field inside a long straight solenoid-carrying current
   A) is zero.
   B) decreases as we move towards its end.
   C) increases as we move towards its end.
   D) is the same at all points.